

As mentioned earlier, the challenge posed by Hadwiger's conjecture is to devise a proof technique that makes better use of the assumption of $\chi \geq r$ than just using its consequence of $\delta \geq r - 1$ in a suitable subgraph, which we know cannot force a K^r minor (Theorem 7.2.4). See the notes for some recent advances in this direction.

If we resign ourselves to using just $\delta \geq r - 1$, we can still ask what additional assumptions might help in making this force a K^r minor. Theorem 7.2.7 says that an assumption of large girth has this effect; see also Exercise 32. In fact, a much weaker assumption suffices: for any fixed $s \in \mathbb{N}$ and all large enough d depending only on s , the graphs $G \not\supseteq K_{s,s}$ of average degree at least d can be shown to have K^r minors for r considerably larger than d . For Hadwiger's conjecture, this implies the following:

Theorem 7.3.8. (Kühn & Osthus 2005)

For every integer s there is an integer r_s such that Hadwiger's conjecture holds for all graphs $G \not\supseteq K_{s,s}$ and $r \geq r_s$.

The strengthening of Hadwiger's conjecture that graphs of chromatic number at least r contain K^r as a *topological* minor has become known as *Hajós's conjecture*. It is false in general, but Theorem 7.2.7 implies it for graphs of large girth:

Corollary 7.3.9. *There is a constant g such that all graphs G of girth at least g satisfy the implication $\chi(G) \geq r \Rightarrow G \supseteq TK^r$ for all r .*

^(5.2.3)
^(7.2.7) *Proof.* Let g be the constant from Theorem 7.2.7. If $\chi(G) \geq r$ then, by Lemma 5.2.3, G has a subgraph H of minimum degree $\delta(H) \geq r - 1$. As $g(H) \geq g(G) \geq g$, Theorem 7.2.7 implies that $G \supseteq H \supseteq TK^r$. $\square$

7.4 Szemerédi's regularity lemma

Some 50 years ago, in the course of the proof of a theorem about arithmetic progressions of integers, Szemerédi developed a graph-theoretical tool that has since come to dominate methods in extremal graph theory like none other: his *regularity lemma*. Very roughly, the lemma says that all graphs can be approximated by random graphs in the following sense: every graph can be partitioned, into a bounded number of equal parts, so that most of its edges run between different parts and the edges between any two parts are distributed fairly uniformly – just as we would expect it if they had been generated at random.

In order to state the regularity lemma precisely, we need some definitions. Let $G = (V, E)$ be a graph, and let $X, Y \subseteq V$ be disjoint. We denote by $\|X, Y\|$ the number of X - Y edges of G , and call

 $\|X, Y\|$

$$d(X, Y) := \frac{\|X, Y\|}{|X||Y|} \quad d(X, Y)$$

the *density* of the pair (X, Y) . (This is a real number between 0 and 1.) Given any $\epsilon > 0$, we call a pair (A, B) of disjoint sets $A, B \subseteq V$ ϵ -regular if all $X \subseteq A$ and $Y \subseteq B$ with

density ϵ -regular pair

$$|X| \geq \epsilon|A| \quad \text{and} \quad |Y| \geq \epsilon|B|$$

satisfy

$$|d(X, Y) - d(A, B)| \leq \epsilon.$$

The edges in an ϵ -regular pair are thus distributed fairly uniformly, the more so the smaller the ϵ we started with.

Consider a partition $\{V_0, V_1, \dots, V_k\}$ of V in which one set V_0 has been singled out as an *exceptional set*. (This exceptional set V_0 may be empty.⁶) We call such a partition an ϵ -regular partition of G if it satisfies the following three conditions:

exceptional set

- (i) $|V_0| \leq \epsilon|V|$;
- (ii) $|V_1| = \dots = |V_k|$;
- (iii) all but at most ϵk^2 of the pairs (V_i, V_j) with $1 \leq i < j \leq k$ are ϵ -regular.

 ϵ -regular partition

The role of the exceptional set V_0 is one of pure convenience: it makes it possible to require that all the other partition sets have exactly the same size. Since condition (iii) affects only the sets $V_1, \dots, V_k$, we may think of V_0 as a kind of bin: its vertices are disregarded when the regularity of the partition is assessed, but there are only few such vertices.

Theorem 7.4.1. (Regularity Lemma)

[7.1.2]
[9.2.2]

For every $\epsilon > 0$ and every integer $m \geq 1$ there exists an integer M such that every graph of order at least m admits an ϵ -regular partition $\{V_0, V_1, \dots, V_k\}$ with $m \leq k \leq M$.

⁶ So V_0 may be an exception also to our terminological rule that partition sets are not normally empty.

The regularity lemma thus says that, given any $\epsilon > 0$, every graph has an ϵ -regular partition into a bounded number of sets. The upper bound M on the number of partition sets ensures that for large graphs the partition sets are large too; note that ϵ -regularity is trivial when the partition sets are singletons, and a powerful property when they are large. The lemma also allows us to specify a lower bound m for the number of partition sets. This can be used to increase the proportion of edges running between different partition sets (i.e., of edges governed by the regularity assertion) over edges inside partition sets (about which we know nothing). See Exercise 39 for an example.

Note that the regularity lemma in this form is designed for use with dense graphs:⁷ for sparse graphs it becomes trivial, because all densities of pairs – and hence their differences – tend to zero (Exercise 40).

The remainder of this section is devoted to the proof of the regularity lemma. Although the proof is not difficult, any reader meeting the regularity lemma here for the first time is likely to draw more insight from seeing how the lemma is typically applied than from studying the technicalities of its proof. Any such reader is encouraged to skip to the start of Section 7.5 now and come back to the proof at his or her leisure.

We shall need the following inequality for reals $\mu_1, \dots, \mu_k > 0$ and $e_1, \dots, e_k \geq 0$:

$$\sum \frac{e_i^2}{\mu_i} \geq \frac{(\sum e_i)^2}{\sum \mu_i}. \quad (1)$$

This follows from the Cauchy-Schwarz inequality $\sum a_i^2 \sum b_i^2 \geq (\sum a_i b_i)^2$ by taking $a_i := \sqrt{\mu_i}$ and $b_i := e_i / \sqrt{\mu_i}$.

$G = (V, E)$
 n Let $G = (V, E)$ be a graph and $n := |V|$. For disjoint sets $A, B \subseteq V$ we define

$$q(A, B) := \frac{|A||B|}{n^2} d^2(A, B) = \frac{\|A, B\|^2}{|A||B|n^2}.$$

For partitions $\mathcal{A}$ of A and $\mathcal{B}$ of B we set

$$q(\mathcal{A}, \mathcal{B}) := \sum_{A' \in \mathcal{A}; B' \in \mathcal{B}} q(A', B'),$$

and for a partition $\mathcal{P} = \{C_1, \dots, C_k\}$ of V we let

$$q(\mathcal{P}) := \sum_{i < j} q(C_i, C_j).$$

⁷ Sparse versions were developed later; see the notes.

However, if $\mathcal{P} = \{C_0, C_1, \dots, C_k\}$ is a partition of V with exceptional set C_0 , we treat C_0 as a set of singletons and define

$$q(\mathcal{P}) := q(\tilde{\mathcal{P}}),$$

where $\tilde{\mathcal{P}} := \{C_1, \dots, C_k\} \cup \{\{v\} : v \in C_0\}$.

 $\tilde{\mathcal{P}}$

The function $q(\mathcal{P})$ plays a pivotal role in our proof of the regularity lemma. On the one hand, it measures the regularity of the partition $\mathcal{P}$: if $\mathcal{P}$ has too many irregular pairs (A, B) , we can take some of the pairs (X, Y) of subsets violating the regularity of the pairs (A, B) and make those sets X and Y into partition sets of their own; as we shall prove, this can refine $\mathcal{P}$ into a partition for which q is substantially greater than for $\mathcal{P}$. Here, ‘substantial’ means that the increase of $q(\mathcal{P})$ is bounded below by some constant depending only on ϵ . On the other hand,

$$\begin{aligned} q(\mathcal{P}) &= \sum_{i < j} q(C_i, C_j) \\ &= \sum_{i < j} \frac{|C_i| |C_j|}{n^2} d^2(C_i, C_j) \\ &\leq \frac{1}{n^2} \sum_{i < j} |C_i| |C_j| \leq 1. \end{aligned}$$

The number of times that $q(\mathcal{P})$ can be increased by a constant is thus also bounded by a constant – in other words, after some bounded number of refinements our partition will be ϵ -regular! To complete the proof of the regularity lemma, all we have to do then is to note how many sets that last partition can possibly have if we start with a partition into m sets, and to choose this number as our promised bound M .

Let us make all this precise. We begin by showing that, when we refine any partition of V , its q -value will not decrease:

Lemma 7.4.2.

- (i) Let $C, D \subseteq V$ be disjoint. If $\mathcal{C}$ is a partition of C and $\mathcal{D}$ is a partition of D , then $q(\mathcal{C}, \mathcal{D}) \geq q(C, D)$.
- (ii) If $\mathcal{P}, \mathcal{P}'$ are partitions of V and $\mathcal{P}'$ refines $\mathcal{P}$, then $q(\mathcal{P}') \geq q(\mathcal{P})$.

Proof. (i) Let $\mathcal{C} = \{C_1, \dots, C_k\}$ and $\mathcal{D} = \{D_1, \dots, D_\ell\}$. Then

$$\begin{aligned} q(\mathcal{C}, \mathcal{D}) &= \sum_{i,j} q(C_i, D_j) \\ &= \frac{1}{n^2} \sum_{i,j} \frac{\|C_i, D_j\|^2}{|C_i| |D_j|} \\ &\stackrel{(1)}{\geq} \frac{1}{n^2} \frac{(\sum_{i,j} \|C_i, D_j\|)^2}{\sum_{i,j} |C_i| |D_j|} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{n^2} \frac{\|C, D\|^2}{(\sum_i |C_i|)(\sum_j |D_j|)} \\
&= q(C, D).
\end{aligned}$$

(ii) Let $\mathcal{P} = \{C_1, \dots, C_k\}$, and for $i = 1, \dots, k$ let $\mathcal{C}_i$ be the partition of C_i induced by $\mathcal{P}'$. Then

$$\begin{aligned}
q(\mathcal{P}) &= \sum_{i < j} q(C_i, C_j) \\
&\leq \sum_{i < j} q(\mathcal{C}_i, \mathcal{C}_j) \\
&\stackrel{(i)}{\leq} q(\mathcal{P}'),
\end{aligned}$$

since $q(\mathcal{P}') = \sum_i q(\mathcal{C}_i) + \sum_{i < j} q(\mathcal{C}_i, \mathcal{C}_j)$. $\square$

Next, we show that by subpartitioning an irregular pair of partition sets we can increase the q -value of that partition a little. Since we are dealing with a single pair only, however, we cannot guarantee yet that the amount of this increase exceeds any constant that depends only on ϵ .

Lemma 7.4.3. *Let $\epsilon > 0$, and let $C, D \subseteq V$ be disjoint. If (C, D) is not ϵ -regular, then there are partitions $\mathcal{C} = \{C_1, C_2\}$ of C and $\mathcal{D} = \{D_1, D_2\}$ of D such that*

$$q(\mathcal{C}, \mathcal{D}) \geq q(C, D) + \epsilon^4 \frac{|C||D|}{n^2}.$$

Proof. Suppose (C, D) is not ϵ -regular. Then there are sets $C_1 \subseteq C$ and $D_1 \subseteq D$ with $|C_1| \geq \epsilon|C|$ and $|D_1| \geq \epsilon|D|$ such that

$$|\eta| > \epsilon \tag{2}$$

for $\eta := d(C_1, D_1) - d(C, D)$. Let $\mathcal{C} := \{C_1, C_2\}$ and $\mathcal{D} := \{D_1, D_2\}$, where $C_2 := C \setminus C_1$ and $D_2 := D \setminus D_1$.

Let us show that $\mathcal{C}$ and $\mathcal{D}$ satisfy the conclusion of the lemma. We shall write $c_i := |C_i|$, $d_i := |D_i|$, $e_{ij} := \|C_i, D_j\|$, $c := |C|$, $d := |D|$ and $e := \|C, D\|$. As in the proof of Lemma 7.4.2,

$$\begin{aligned}
q(\mathcal{C}, \mathcal{D}) &= \frac{1}{n^2} \sum_{i,j} \frac{e_{ij}^2}{c_i d_j} \\
&= \frac{1}{n^2} \left(\frac{e_{11}^2}{c_1 d_1} + \sum_{i+j > 2} \frac{e_{ij}^2}{c_i d_j} \right) \\
&\stackrel{(1)}{\geq} \frac{1}{n^2} \left(\frac{e_{11}^2}{c_1 d_1} + \frac{(e - e_{11})^2}{cd - c_1 d_1} \right).
\end{aligned}$$

By definition of η , we have $e_{11} = c_1 d_1 e / cd + \eta c_1 d_1$, so

$$\begin{aligned}
 n^2 q(\mathcal{C}, \mathcal{D}) &\geq \frac{1}{c_1 d_1} \left(\frac{c_1 d_1 e}{cd} + \eta c_1 d_1 \right)^2 \\
 &\quad + \frac{1}{cd - c_1 d_1} \left(\frac{cd - c_1 d_1}{cd} e - \eta c_1 d_1 \right)^2 \\
 &= \frac{c_1 d_1 e^2}{c^2 d^2} + \frac{2e\eta c_1 d_1}{cd} + \eta^2 c_1 d_1 \\
 &\quad + \frac{cd - c_1 d_1}{c^2 d^2} e^2 - \frac{2e\eta c_1 d_1}{cd} + \frac{\eta^2 c_1^2 d_1^2}{cd - c_1 d_1} \\
 &\geq \frac{e^2}{cd} + \eta^2 c_1 d_1 \\
 &\stackrel{(2)}{\geq} \frac{e^2}{cd} + \epsilon^4 cd = n^2 q(C, D) + \epsilon^4 |C| |D|
 \end{aligned}$$

as claimed, since $c_1 \geq \epsilon c$ and $d_1 \geq \epsilon d$ by the choice of C_1 and D_1 . $\square$

Finally, we show that if a partition has enough irregular pairs of partition sets to fall short of the definition of an ϵ -regular partition, then subpartitioning all those pairs at once can increase q by a constant:

Lemma 7.4.4. *Let $0 < \epsilon \leq 1/4$, and let $\mathcal{P} = \{C_0, C_1, \dots, C_k\}$ be a partition of V , with exceptional set C_0 of size $|C_0| \leq \epsilon n$ and $|C_1| = \dots = |C_k| =: c$. If $\mathcal{P}$ is not ϵ -regular, then there is a partition $\mathcal{P}' = \{C'_0, C'_1, \dots, C'_\ell\}$ of V with exceptional set C'_0 , where $k \leq \ell \leq k4^{k+1}$, such that $|C'_0| \leq |C_0| + n/2^k$, all other sets C'_i have equal size, and either $\mathcal{P}'$ is ϵ -regular or*

$$q(\mathcal{P}') \geq q(\mathcal{P}) + \epsilon^5/2.$$

Proof. For all $1 \leq i < j \leq k$, let us define a partition $\mathcal{C}_{ij}$ of C_i and a partition $\mathcal{C}_{ji}$ of C_j , as follows. If the pair (C_i, C_j) is ϵ -regular, we let $\mathcal{C}_{ij} := \{C_i\}$ and $\mathcal{C}_{ji} := \{C_j\}$. If not, then by Lemma 7.4.3 there are partitions $\mathcal{C}_{ij}$ of C_i and $\mathcal{C}_{ji}$ of C_j with $|\mathcal{C}_{ij}| = |\mathcal{C}_{ji}| = 2$ and

$$q(\mathcal{C}_{ij}, \mathcal{C}_{ji}) \geq q(C_i, C_j) + \epsilon^4 \frac{|C_i| |C_j|}{n^2} = q(C_i, C_j) + \frac{\epsilon^4 c^2}{n^2}. \quad (3)$$

For each $i = 1, \dots, k$, let $\mathcal{C}_i$ be the unique minimal partition of C_i that refines every partition $\mathcal{C}_{ij}$ with $j \neq i$. (In other words, if we consider two elements of C_i as equivalent whenever they lie in the same partition set

of $\mathcal{C}_{ij}$ for every $j \neq i$, then $\mathcal{C}_i$ is the set of equivalence classes.) Thus, $|\mathcal{C}_i| \leq 2^{k-1}$. Now consider the partition

$$\mathcal{C} := \{C_0\} \cup \bigcup_{i=1}^k \mathcal{C}_i$$

of V , with C_0 as exceptional set. Then $\mathcal{C}$ refines $\mathcal{P}$ and $|\mathcal{C} \setminus \{C_0\}| \leq k2^{k-1}$, so

$$k \leq |\mathcal{C}| \leq k2^k. \quad (4)$$

Let $\mathcal{C}_0 := \{\{v\} \mid v \in C_0\}$. Now if $\mathcal{P}$ is not ϵ -regular, then for more than ϵk^2 of the pairs (C_i, C_j) with $1 \leq i < j \leq k$ the partition $\mathcal{C}_{ij}$ is non-trivial. Hence, by our definition of q for partitions with exceptional set, and Lemma 7.4.2 (i),

$$\begin{aligned} q(\mathcal{C}) &= \sum_{1 \leq i < j} q(\mathcal{C}_i, \mathcal{C}_j) + \sum_{1 \leq i} q(\mathcal{C}_0, \mathcal{C}_i) + \sum_{0 \leq i} q(\mathcal{C}_i) \\ &\geq \sum_{1 \leq i < j} q(\mathcal{C}_{ij}, \mathcal{C}_{ji}) + \sum_{1 \leq i} q(\mathcal{C}_0, \{C_i\}) + q(\mathcal{C}_0) \\ &\geq \sum_{1 \leq i < j} q(C_i, C_j) + \epsilon k^2 \frac{\epsilon^4 c^2}{n^2} + \sum_{1 \leq i} q(\mathcal{C}_0, \{C_i\}) + q(\mathcal{C}_0) \\ &\quad (3) \\ &= q(\mathcal{P}) + \epsilon^5 \left(\frac{kc}{n} \right)^2 \\ &\geq q(\mathcal{P}) + \epsilon^5/2. \end{aligned}$$

(For the last inequality, recall that $|C_0| \leq \epsilon n \leq \frac{1}{4}n$, so $kc \geq \frac{3}{4}n$.)

In order to turn $\mathcal{C}$ into our desired partition $\mathcal{P}'$, all that remains to do is to cut its sets up into pieces of some common size so that all remaining vertices can be collected into the exceptional set without making this too large.

If $c < 4^k$, the ϵ -regular partition $\mathcal{P}'$ into $C'_0 := C_0$ and the singletons $\{v\}$ with $v \in V \setminus C_0$ is as desired, since there are ℓ such singletons for $k \leq \ell = kc < k4^k$.

Assume now that $c \geq 4^k$. Let $C'_1, \dots, C'_\ell$ be a maximal collection of disjoint sets of size $d := \lfloor c/4^k \rfloor \geq 1$ such that each C'_i is contained in some $C \in \mathcal{C} \setminus \{C_0\}$, and put $C'_0 := V \setminus \bigcup C'_i$. Then $\mathcal{P}' = \{C'_0, C'_1, \dots, C'_\ell\}$ is indeed a partition of V . Moreover, $\tilde{\mathcal{P}}'$ refines $\tilde{\mathcal{C}}$, so

$$q(\mathcal{P}') \geq q(\mathcal{C}) \geq q(\mathcal{P}) + \epsilon^5/2$$

by Lemma 7.4.2 (ii). Since each set $C'_i \neq C'_0$ is also contained in one of the sets $C_1, \dots, C_k$, but no more than $c/d \leq 4^{k+1}$ sets C'_i can lie inside

the same C_j (by the choice of d), we also have $k \leq \ell \leq k4^{k+1}$ as required. Finally, the sets $C'_1, \dots, C'_\ell$ use all but at most d vertices from each set $C \neq C_0$ of $\mathcal{C}$. Hence,

$$\begin{aligned} |C'_0| &\leq |C_0| + d|\mathcal{C}| \\ &\stackrel{(4)}{\leq} |C_0| + \frac{c}{4^k} k 2^k \\ &= |C_0| + ck/2^k \\ &\leq |C_0| + n/2^k. \end{aligned}$$

□

We are not ready to prove the regularity lemma. Our plan is, roughly, to start for fixed ϵ with any partition $\mathcal{P}$ of V to which we can apply Lemma 7.4.4. We shall iterate this until the partition the lemma returns is ϵ -regular. This should happen after at most $2/\epsilon^5$ iterations, since the q -value of our partition grows by at least $\epsilon^5/2$ at each step but cannot exceed 1. The number of sets in our final partition will therefore also be bounded in terms of ϵ , and we can take this bound as the value for M required in the assertion of the regularity lemma.

Proof of Theorem 7.4.1. Let $\epsilon > 0$ and $m \geq 1$ be given, and assume without loss of generality that $\epsilon \leq 1/4$. Let $s := 2/\epsilon^5$.

 ϵ, m

There is one critical requirement which a partition $\{C_0, C_1, \dots, C_k\}$ with $|C_1| = \dots = |C_k|$ has to satisfy before Lemma 7.4.4 can be (re-)applied: the size $|C_0|$ of its exceptional set must not exceed ϵn . As the size of the exceptional set can grow with each iteration of the lemma, see there, we thus have to start with C_0 small enough that after up to $s - 1$ increases its size is still less than ϵn .

The increase to $|C_0|$ effected by one application of Lemma 7.4.4 is stated there as being at most $n/2^k$. This k is the number of non-exceptional sets in the partition to which we apply the lemma, so $n/2^k$ decreases with each step. Hence we only need to start with k large enough that s increments even of our starting value of $n/2^k$ add up to at most $\epsilon n/2$ and ensure, if we can, that the initial size of C_0 is also at most $\epsilon n/2$.

Pick $k \geq m$ large enough that $2^{k-1} \geq s/\epsilon$. Then $s/2^k \leq \epsilon/2$, and hence

 k

$$k + sn/2^k \leq \epsilon n \tag{5}$$

whenever $k \leq \epsilon n/2$, i.e. for all $n \geq 2k/\epsilon$.

Let us now choose M . This should be an upper bound on the number of non-exceptional sets in our partition after up to s iterations of Lemma 7.4.4, where in each iteration this number may grow from its current value r to at most $r4^{r+1}$. So let f be the function $x \mapsto x4^{x+1}$,

and take $M := \max \{f^s(k), 2k/\epsilon\}$; the second term in the maximum ensures that any $n \geq M$ is large enough to satisfy (5).

We finally have to show that every graph $G = (V, E)$ of order at least m has an ϵ -regular partition $\{V_0, V_1, \dots, V_{k'}\}$ with $m \leq k' \leq M$. So let G be given, and let $n := |G|$. If $n \leq M$, we partition G into $k' := n$ singletons, choosing $V_0 := \emptyset$ and $|V_1| = \dots = |V_{k'}| = 1$. This partition of G is clearly ϵ -regular. Suppose now that $n > M$. Let $C_0 \subseteq V$ be minimal such that k divides $|V \setminus C_0|$, and let $\{C_1, \dots, C_k\}$ be any partition of $V \setminus C_0$ into sets of equal size. Then $|C_0| \leq k - 1$, and hence $|C_0| \leq \epsilon n$ by (5). Starting with $\{C_0, C_1, \dots, C_k\}$ we apply Lemma 7.4.4 again and again, until the partition of G obtained is ϵ -regular; this will happen after at most s iterations, since by (5) the size of the exceptional set in the partitions stays below ϵn , so the lemma could indeed be reapplied up to the theoretical maximum of s times. $\square$

7.5 Applying the regularity lemma

The purpose of this section is to illustrate how the regularity lemma is typically applied in the context of (dense) extremal graph theory. Suppose we are trying to prove that a certain edge density of a graph G suffices to force the occurrence of some given subgraph H , and that we have an ϵ -regular partition of G . For most of the pairs (V_i, V_j) of partition sets, the edges between V_i and V_j are distributed fairly uniformly; their density, however, may depend on the pair. But since G has many edges, this density cannot be too small for too many pairs: some sizeable proportion of the pairs will have at least a certain positive density. Moreover if G is large, then so are the pairs: recall that the number of partition sets is bounded, and they have equal size. But any large enough bipartite graph with equal partition sets, fixed positive edge density (however small) and a uniform distribution of edges will contain any given bipartite subgraph;⁸ this will be made precise below. Writing H as a union of bipartite subgraphs, say those induced by pairs of colour classes of some vertex colouring of H , we shall obtain $H \subseteq G$ as desired.

These ideas will be formalized by the so-called ‘blow-up lemma’, Lemma 7.5.2 below. We shall then use this and the regularity lemma to prove the Erdős-Stone theorem from Section 7.1; another application will be given later, in the proof of Theorem 9.2.2. We wind up the section with an informal review of the application of the regularity lemma that we have seen, summarizing what it can teach us for similar applications. In particular, we look at how the various parameters involved depend

⁸ Readers already acquainted with random graphs may find it instructive to compare this statement with Proposition 11.3.1.